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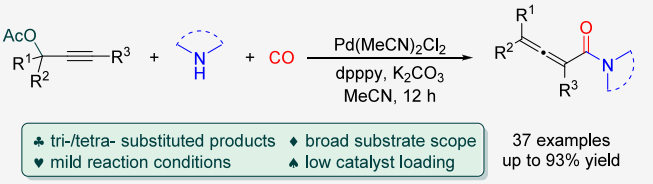
Synthesis of Multisubstituted 2,3-Allenamides via Palladium-Catalyzed Carbonylation of Propargylic Esters

Zhi Zhang, Miao-Miao Ji, Xiao-Feng Wu, Yong-Yu He, and Jin-Bao Peng*

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ABSTRACT: 2,3-Allenamides are an important class of unsaturated group-substituted carbonyl compounds. A palladium-catalyzed aminocarbonylation of propargyl acetates with amines for the synthesized tri-/tetrasubstituted 2,3-allenamides has been developed. A broad range of tri-/tetrasubstituted 2,3-allenamides have been prepared from propargyl acetates in good to excellent yields. The reaction featured mild reaction conditions and good functional group tolerance. The applicability of this methodology was further highlighted by the late-stage modification of several natural products and pharmaceuticals.



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INTRODUCTION

2,3-Allenamides are an important class of unsaturated group-substituted carbonyl compounds.[1] This type of structure exists in natural products, medicinal chemistry, and agrochemical industry (Figure 1).[2] For example, allenamide compound III was used to selectively label target cysteine residues in fully unprotected peptides and proteins.[2c] Compound IV was reported to inhibit T790M/L858R double mutant and wild type EGFR kinases in a low nanomolar range.[2] In addition, 2,3-allenamides also serve as versatile building blocks in organic synthesis due to their unique structure and reactivity. With a unique orthogonal cumulative π-system, this structure has been widely used in a range of transformations, such as cyclization reactions,[3] nucleophilic and radical addition reactions,[4] and cycloaddition reactions.[5]

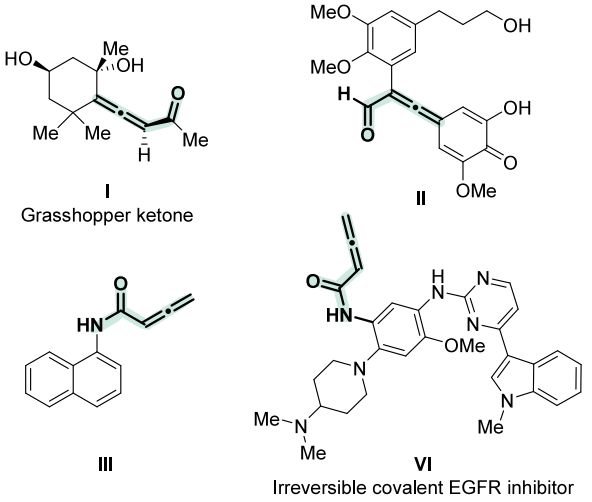
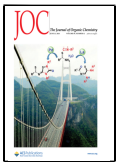


Figure 1. Representative natural products and bioactive molecules containing related skeletons.

Traditional methods for the synthesis of allenoates and 2,3-allenoic acids is based on a Wittig-type reaction,[6] which usually yielded a mixture of allenoates and alkynoates. Direct carboxylation reaction of allenylmetal reagents with carbon dioxide was also reported,[7] but this method requires preparation of organometallic reagents and suffers from poor functional group tolerance. Many other strategies have also been developed for the synthesis of allenyl ketones or allenoates, such as transition metal catalyzed β-H elimination of enol triflates[8] and 1,4-difunctionalization of enyne compounds.[9] Although there are many synthetic studies on allenoic acids, allenoic ketones, and allenoates, there are only limited reports on the synthesis of 2,3-allenamides.

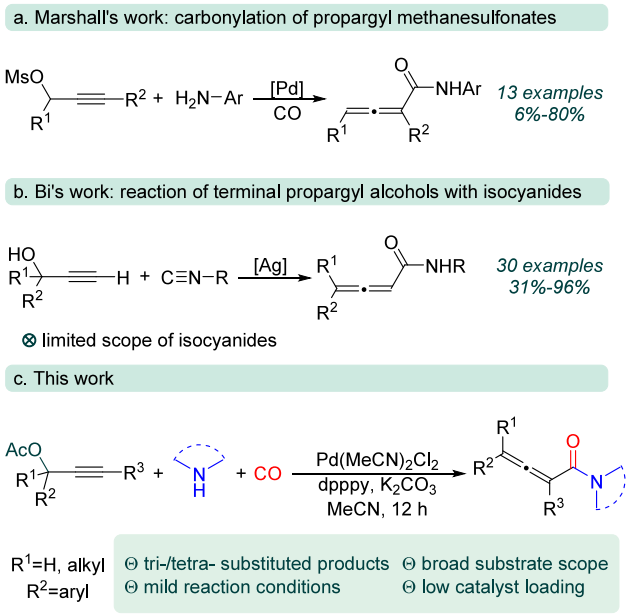
On the contrary, the transition-metal-catalyzed carbonylation reaction has become one of the most powerful tools for the synthesis of carbonyl compounds.[10] The carboxylation of the readily available propargylic alcohols and their derivatives with CO provides an efficient way for synthesis of 2,3-allenoic compounds.[11] Marshall reported the synthesis of trisubstituted 2,3-allenamide compounds from propargyl methanesulfonate. The reaction requires high CO pressure to achieve good selectivity; otherwise, a large amount of propargyl amines will be produced (Scheme 1a).[12] In 2014, Bi and co-workers reported a silver-catalyzed cross-coupling of propargylic alcohols with isocyanides for the synthesis of di/

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Scheme 1. Synthesis of 2,3-Allenamides from Propargyl Compounds



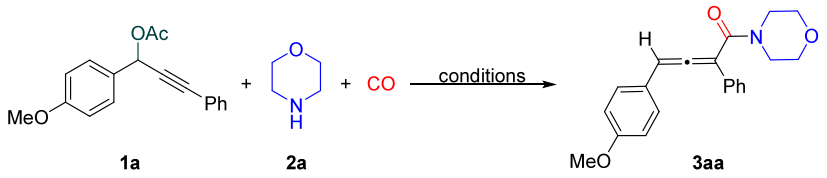
trisubstituted 2,3-allenamides (Scheme 1b).[13] However, only two primary isocyanides were reported to undergo clean conversion. The direct carbonylation of propargyl alcohol derivatives and secondary aliphatic amine feedstocks to synthesize 2,3-allenamides is rare.[14] Thus, the development of a general and effective method for the synthesis of 2,3-allenamides is still highly desirable.

With a continuous interest in the reactions involving propargyl alcohol and the carbonylation reaction,[15] we herein report a palladium-catalyzed aminocarbonylation of propargyl acetates with amines for the synthesize tri-/tetrasubstituted 2,3-allenamides (Scheme 1c). The main features of this reaction include the following: (1) preparation of tri-/tetrasubstituted 2,3-allenamides with propargyl acetates under mild reaction conditions, (2) broad substrate scopes (including some natural products and pharmaceutical intermediates) as well as good functional group tolerance, and (3) efficient catalysis with low catalyst loading.

RESULTS AND DISCUSSION

Our investigation commenced with 1-(4-methoxyphenyl)-3-phenylprop-2-yn-1-yl acetate (**1a**) and morpholine (**2a**) as model substrates. As shown in Table 1, in the presence of $Pd(CH_3CN)_2Cl_2$, PPh_3, and quinine in CH_3CN in the presence of CO at 60 °C for 12 h, the reaction furnished the expected 2,3-allenamide (**3aa**) in 46% yield (entry 1). Notably, the in situ generated acetic formic anhydride was used as an CO surrogate.[16] First, we screened different inorganic bases and organic bases (entries 1−4). When K_2CO_3 was utilized, the reaction yield was significantly improved (61%). Then, various palladium catalysts were investigated (entries 4−7), and the results showed that $Pd(CH_3CN)_2Cl_2$ is optimal. Investigation of the reaction temperatures (entries 7−9) shows that the reaction temperature of 50 °C (entry 9) is suitable, and the product **3aa** was obtained in 70% yield. Lower conversion was observed at lower temperatures (entry 8). The amount of base is also important for the reaction (entries 9−11). After optimization, it was found that 1.0 equiv of K_2CO_3 was suitable, and the reaction yield was 76% (entry 11). Subsequently, the concentration of the reaction system was examined (entries 11−13), and the results showed that the solvent amount of 2.0 mL has a better yield (entry 13, 83%).

Table 1. Optimization of the Reaction Parameters[a]



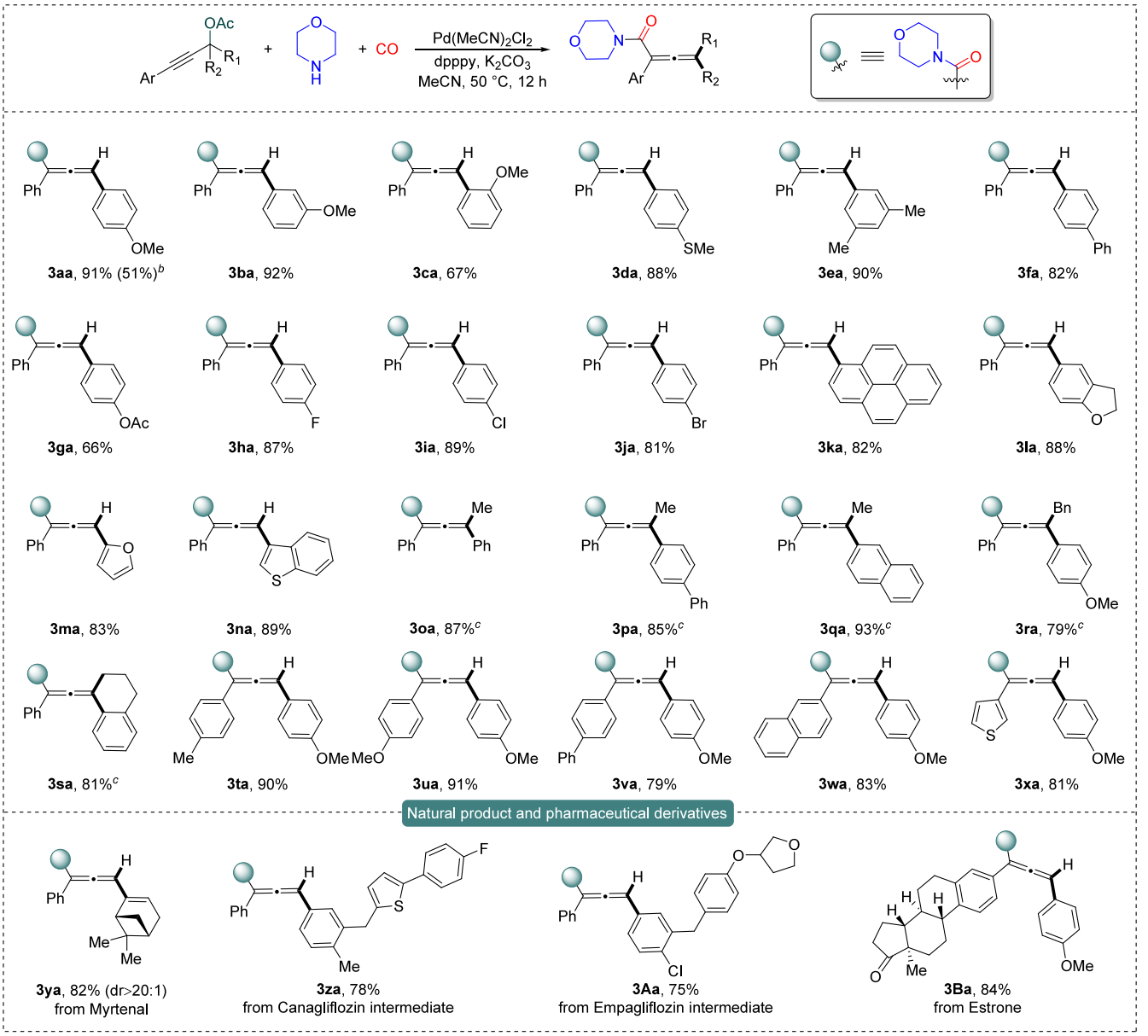
1a **2a** **3aa**

Entry	Base	[Pd]	Temp. (°C)	Ligand	Yield (%)[b]
1	quinine	$Pd(MeCN)_2Cl_2$	60	PPh_3	46
2	pyridine	$Pd(MeCN)_2Cl_2$	60	PPh_3	45
3	Cs_2CO_3	$Pd(MeCN)_2Cl_2$	60	PPh_3	30
4	K_2CO_3	$Pd(MeCN)_2Cl_2$	60	PPh_3	61
5	K_2CO_3	$Pd(OAc)_2$	60	PPh_3	52
6	K_2CO_3	$Pd(PPh_3)_4$	60	PPh_3	42
7	K_2CO_3	$PdCl_2$	60	PPh_3	28
8	K_2CO_3	$Pd(MeCN)_2Cl_2$	40	PPh_3	37
9	K_2CO_3	$Pd(MeCN)_2Cl_2$	50	PPh_3	70
10[c]	K_2CO_3	$Pd(MeCN)_2Cl_2$	50	PPh_3	55
11[d]	K_2CO_3	$Pd(MeCN)_2Cl_2$	50	PPh_3	76
12[e]	K_2CO_3	$Pd(MeCN)_2Cl_2$	50	PPh_3	61
13[f]	K_2CO_3	$Pd(MeCN)_2Cl_2$	50	PPh_3	83
14	K_2CO_3	$Pd(MeCN)_2Cl_2$	50	dppb	trace
15	K_2CO_3	$Pd(MeCN)_2Cl_2$	50	PCy_3	59
16	**K_2CO_3**	**$Pd(MeCN)_2Cl_2$**	**50**	**dpppy**	**91**

[a]Reaction conditions: **1** (0.2 mmol), **2** (0.3 mmol), [Pd] (2 mol %), ligand (5 mol % for monodentate ligands, 2.5 mol % for bidentate ligands), base (0.3 mmol), solvent (1.0 mL), [CO] (HCO_2H + Ac_2O, 2 mmol), *T* °C for 12 h. [b]Isolated yield. [c]K_2CO_3 (0.1 mmol). [d]K_2CO_3 (0.2 mmol). [e]MeCN (3.0 mL). [f]MeCN (2.0 mL). dpppy = diphenyl-2-pyridylphosphine.

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Scheme 2. Substrate Scope of Propargyl Alcohol Acetate[a]



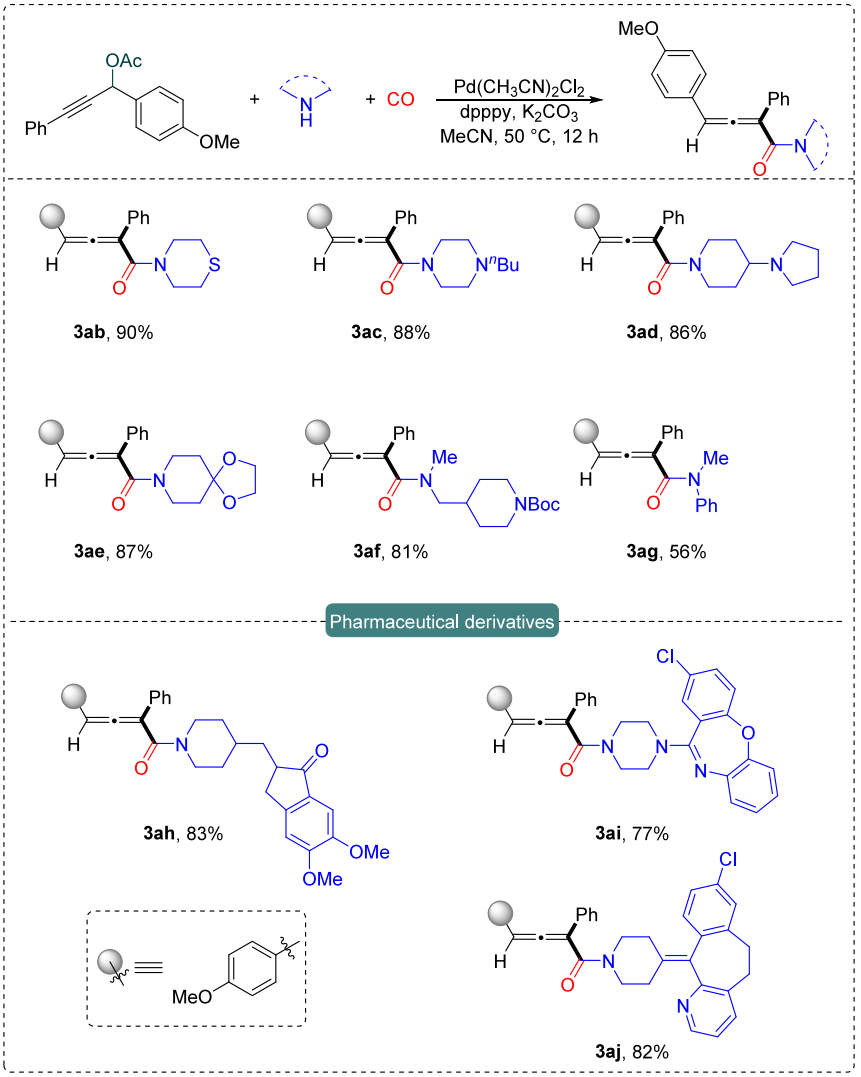
[a]Reaction conditions: **1** (0.2 mmol), **2** (0.3 mmol), $Pd(MeCN)_2Cl_2$ (2 mol %), dpppy (5 mol %), K_2CO_3 (0.2 mmol), MeCN (2.0 mL), [CO] (HCO_2H + Ac_2O, 2 mmol), 50 °C for 12 h. [b]**1a** (1.0 mmol), **2a** (1.5 mmol). [c]60 °C.

Finally, we examined the effect of the ligands (entries 13−16). Bidentate ligands were less effective in this reaction (entry 14, and see details in Supporting Information (SI)). Among the monodentate ligands, dpppy has a good effect on the reaction (entry 16, 91%).

With the optimal reaction conditions in hand, the scope of the method was examined. As shown in Scheme 2, a variety of propargyl acetates bearing electron-neutral, electron-rich, and electron-deficient aryl moieties underwent this transformation smoothly, forming tri-/tetrasubstituted 2,3-allenamides. A variety of functional groups, including OMe, SMe, Me, Ph, OAc, F, Cl, and Br, on the aromatic ring of the substrates were all well tolerated (**3aa**−**3ja**, 67−92%), with the halogen moiety acting as a potential synthetic handle for subsequent modification (**3ha**−**3ja**). It was noted that the steric hindrance has an influence on the reactivity: the high steric hindrance was detrimental to the reactivity (**3aa**, 91% vs **3ca**, 67%). Products with electron-donating and -withdrawing groups (**3aa**−**3ja**, 66−92%) can be successfully prepared. Notably, 4-(1-acetoxy-3-phenylprop-2-yn-1-yl)phenyl acetate (**1g**) was suitable for this reaction and gave allenamide **3ga** (66%). Several useful aromatic functionalities, such as pyrene-1-yl (**1k**), 2,3-dihydrobenzofuran-5-yl (**1l**), furan-2-yl (**1m**), and benzo[*b*]-thiophene-3-yl (**1n**), accommodated the aminocarbonylation furnishing **3ka**−**3na** in 82−89% yields. Then, we attempted to synthesize tetrasubstituted allenamide compounds (**3oa**−**3sa**) via tertiary propargyl acetates (Scheme 2). The results show that the reaction temperature of 60 °C is more conducive to the production of the target product. Propargyl acetates **1o**−**1s** were efficiently converted to **3oa**−**3sa** in good to excellent yields (79−93%), and it is noteworthy that terminal cyclic tertiary propargyl acetate **1s** was tolerated to furnish product

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Scheme 3. Substrate Scope of Secondary Amines[a]



[a]Reaction conditions: **1** (0.2 mmol), **2** (0.3 mmol), $Pd(MeCN)_2Cl_2$ (2 mol %), dpppy (5 mol %), K_2CO_3 (0.2 mmol), MeCN (2.0 mL), [CO] (HCO_2H + Ac_2O, 2 mmol), 50 °C for 12 h.

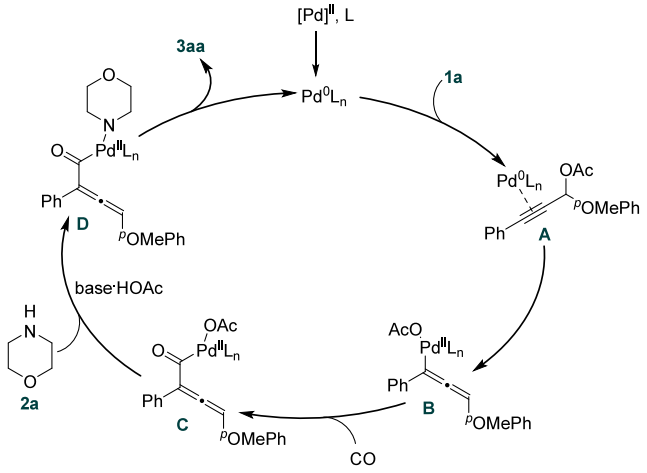
3sa (81%). We also investigated the terminal aromatic functionalities of propargyl acetates (Scheme 2). When the terminal aromatic functionalities are *p*-tolyl (**3ta**), *p*-methoxyphenyl (**3ua**), biphenyl (**3va**) or naphthyl (**3wa**), and thienyl (**3xa**), the product yields are 79−91%. In addition, in order to verify the applicability of this method, we synthesized the corresponding 2,3-allenamide derivatives from natural products and pharmaceutical intermediates. Compound **3ya** (82%) derived from (−)-myrtenal. The intermediates of canagliflozin and empagliflozin can be successfully synthesized into their derivatives **3za**−**3Aa** (75−78%) and compound **3Ba** (84%) derived from the natural product estrone.

Finally, we expanded the scope of amine compounds using 1-(4-methoxyphenyl)-3-phenylprop-2-yn-1-yl acetate (Scheme 3). We found that a variety of secondary amines are suitable for this reaction. Common drug fragments such as thiomorpholine, piperazine, and piperidine can be reacted smoothly, and the products **3ab**−**3ad** are obtained in excellent yields (86−90%). The ketal compound 4-piperidone vinyl ketal can successfully prepare compound **3ae** (87%). Acyclic amine **2e** containing amide and ester structures can react smoothly to obtain compound **3af** (81%). We also prepared some drug derivatives based on drugs or drug fragments. The donepezil derivative **3ag** (82%) can be obtained by using debenzyldonepezil. Desloratadine and amoxapine can also be used directly to synthesize drug derivatives **3ah** (82%) and **3ai** (77%). In addition, we investigated the reactivity of the aniline compounds in this reaction. When *N*-methylaniline was used to react under standard reaction conditions, the target compound **3aj** could be obtained in moderate yield (56%).

Based on our research and previous related work,[14b] we proposed a possible mechanism (Scheme 4). Taking the generation of **3aa** as an example, the initially situ-generated active Pd(0) species performs ligand exchange with propargyl alcohol **1a** acetate to give the precursor **A**, which affords allenylic palladium intermediate **B** through oxidative addition. Subsequently, coordination and migratary insertion of CO formed acylpalladium intermediate **C**. Next, ligand exchange of acetate with amine in the presence of base generated intermediate **D**, which undergoes reductive elimination to give product **3aa**. Meanwhile, Pd(II) is converted to Pd(0) and participates in the next catalytic cycle.

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Scheme 4. A Plausible Reaction Pathway



CONCLUSIONS

In summary, we have developed a novel palladium-catalyzed aminocarbonylation of propargyl acetates for the one-step synthesis of 2,3-allenamide compounds. The transformation uses easily available substrates to efficiently deliver a broad range of multisubstituted 2,3-allenamides under mild conditions and low pressures of CO. This protocol exhibits broad substrate scope as well as good functional group tolerance and facile capacity of modification of natural products and pharmaceuticals.

EXPERIMENTAL SECTION

General Information. Unless otherwise noted, all reactions were carried out under a nitrogen atmosphere. All reagents were obtained from commercial sources and used as received without further purification. All solvents were dried by standard techniques and distilled prior to use. Column chromatography was performed on silica gel (200−300 meshes) using petroleum ether (bp 60−90 °C) and ethyl acetate as eluent. Column chromatography was performed on neutral alumina (200−300 meshes) using petroleum ether (bp 60−90 °C) and ethyl acetate as eluent. ^{1}NMR spectra were recorded on a Bruker Avance operating at for ^{1}H NMR at 500 MHz, ^{13}C NMR at 126 MHz, and ^{19}F NMR at 471 MHz, and spectral data were reported in ppm relative to tetramethylsilane (TMS) as the internal standard and $CDCl_3$ (^{1}H NMR δ 7.26, ^{13}C NMR δ 77.16) as the solvent. High-resolution mass spectra (HRMS) are produced by Thermo Fisher Scientific. Its main body is composed of two parts: Thermo Scientific's UltiMate 3000 Series liquid system and Thermo Scientific Q-Exactive combined quadrupole Orbitrap mass spectrometer. All coupling constants (J) are reported in Hz. The following abbreviations were used to describe peak splitting patterns when appropriate: s = singlet, d = doublet, dd = double doublet, ddd = double doublet of doublets, t = triplet, dt = double triplet, q = quatriplet, m = multiplet, and br = broad. Compounds **1a−1B** were prepared according to the previous literature (refs S1−S6 in the SI). All the amines used here were commercially available. Amines **2a**, **2e**, **2g** were purchased from Energy Chemical, and compounds **2b−2d**, **2f**, **2h−2j** were purchased from Bide Pharmatech and Macklin.

General Procedure. **1** (0.2 mmol, 1 equiv), $Pd(MeCN)_2Cl_2$ (1.0 mg, 2 mol %), dpppy (2.6 mg, 5 mol %), and K_2CO_3 (27.6 mg, 1.0 equiv) were transferred into a 15 mL tube. A 2.0 mL vial was placed in the tube, and the tube was sealed with a septum. The tube was connected to a nitrogen-vacuum line, evacuated, and backfilled with N_2 (×3). Then, a solution of **2** (0.3 mmol, 1.5 equiv) in MeCN (2.0 mL) was added to the reaction tube. To the inner vial were added Et_3N (276 μL, 2.0 mmol) and HCO_2H/Ac_2O (which was premixed and stirred for 30 min, 75 μL/187 μL, 2.0 mmol). Then the reaction tube was sealed with a screw-top septum cap quickly and placed in a heating block that was preheated to 50/60 °C. After a time period of 12 h, the reaction tube was allowed to cool to room temperature. The mixture was concentrated under reduced pressure, and the residue was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) to afford the products **3**.

Experimental Characterization Data of Products. *4-(4-Methoxyphenyl)-1-morpholino-2-phenylbuta-2,3-dien-1-one (**3aa**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a yellow solid (61.0 mg, 91% yield, mp 125.7−127.1 °C). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.42−7.39 (m, 2H), 7.35 (dd, J = 10.4, 4.9 Hz, 2H), 7.30−7.25 (m, 3H), 6.89−6.85 (m, 2H), 6.73 (s, 1H), 3.87−3.80 (m, 4H), 3.73 (dd, J = 12.4, 8.1 Hz, 3H), 3.64−3.49 (m, 4H). ^{13}C{^{1}H} NMR (126 MHz, $CDCl_3$) δ: 203.8, 165.1, 159.7, 132.3, 129.1, 128.7, 128.2, 126.5, 124.7, 114.6, 107.5, 100.4, 67.1, 67.0, 55.5, 47.81, 42.5. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{21}H_{22}NO_3^+$ 336.1597; Found 336.1588.

*5-(3-Methoxyphenyl)-1-morpholino-2-phenylbuta-2,3-dien-1-one (**3ba**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a yellow solid (61.6 mg, 92% yield, mp 111.7−113.5 °C). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.41 (d, J = 7.5 Hz, 2H), 7.35 (t, J = 7.6 Hz, 2H), 7.27 (t, J = 7.3 Hz, 3H), 6.87 (d, J = 8.6 Hz, 2H), 6.73 (s, 1H), 3.88−3.78 (m, 4H), 3.73 (dd, J = 12.0, 7.9 Hz, 3H), 3.64−3.50 (m, 4H). ^{13}C{^{1}H} NMR (126 MHz, $CDCl_3$) δ: 203.8, 165.0, 159.7, 132.3, 129.1, 128.7, 128.1, 126.5, 124.7, 114.6, 107.5, 100.4, 67.0, 66.9, 55.5, 47.8, 42.5. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{21}H_{22}NO_3^+$ 336.1597; Found 336.1588.

*4-(2-Methoxyphenyl)-1-morpholino-2-phenylbuta-2,3-dien-1-one (**3ca**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a colorless liquid (44.9 mg, 67% yield). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.43 (d, J = 7.5 Hz, 2H), 7.38−7.32 (m, 3H), 7.27 (d, J = 7.8 Hz, 1H), 7.25−7.22 (m, 1H), 7.10 (s, 1H), 6.91 (t, J = 7.2 Hz, 2H), 3.84 (d, J = 11.5 Hz, 4H), 3.73 (dt, J = 13.8, 9.2 Hz, 3H), 3.65−3.48 (m, 4H). ^{13}C{^{1}H} NMR (126 MHz, $CDCl_3$) δ: 205.1, 165.1, 156.6, 132.6, 129.4, 129.0, 128.9, 127.9, 126.6, 121.1, 121.0, 111.2, 106.4, 95.2, 67.1, 67.0, 55.7, 47.7, 42.5. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{21}H_{22}NO_3^+$ 336.1597; Found 336.1588.

*4-(4-(Methylthio)phenyl)-1-morpholino-2-phenylbuta-2,3-dien-1-one (**3da**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a yellow liquid (61.8 mg, 88% yield). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.41−7.38 (m, 2H), 7.37−7.33 (m, 2H), 7.28 (dd, J = 11.1, 3.9 Hz, 3H), 7.22−7.19 (m, 2H), 6.72 (s, 1H), 3.87−3.80 (m, 1H), 3.77−3.70 (m, 3H), 3.61 (d, J = 4.3 Hz, 2H), 3.58−3.56 (m, 1H), 3.55−3.50 (m, 1H), 2.48 (s, 3H). ^{13}C{^{1}H} NMR (126 MHz, $CDCl_3$) δ: 204.3, 164.8, 138.8, 132.1, 129.2, 128.3, 127.9, 127.0, 126.6, 107.8, 100.5, 67.1, 67.0, 47.8, 42.5, 15.9. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{21}H_{22}NO_2S^+$ 352.1369; Found 352.1360.

*4-(3,5-Dimethylphenyl)-1-morpholino-2-phenylbuta-2,3-dien-1-one (**3ea**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a yellow liquid (59.9 mg, 90% yield). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.42 (d, J = 7.4 Hz, 2H), 7.36 (t, J = 7.6 Hz, 2H), 7.31−7.25 (m, 1H), 6.97 (s, 2H), 6.92 (s, 1H), 6.71 (s, 1H), 3.91−3.83 (m, 1H), 3.79−3.58 (m, 6H), 3.56−3.51 (m, 1H), 2.29 (s, 6H). ^{13}C{^{1}H} NMR (126 MHz, $CDCl_3$) δ: 204.3, 164.9, 138.7, 132.3, 132.2, 130.0, 129.1, 128.2, 126.6, 125.3, 107.3, 101.0, 67.0, 66.9, 47.8, 42.5, 21.4. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{22}H_{24}NO_2^+$ 334.1805; Found 334.1793.

*4-([1,1′-Biphenyl]-4-yl)-1-morpholino-2-phenylbuta-2,3-dien-1-one (**3fa**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a white solid (62.5 mg, 82% yield, mp 88.8−89.6 °C). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.60−7.56 (m, 4H), 7.45 (dd, J = 11.5, 5.0 Hz, 6H), 7.40−7.33 (m, 3H),

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7.29 (t, J = 7.3 Hz, 1H), 6.81 (s, 1H), 3.88−3.81 (m, 1H), 3.75 (dd, J = 10.1, 5.0 Hz, 3H), 3.68−3.52 (m, 4H). $^{13}C\{^1H\}$ NMR (126 MHz, $CDCl_3$) δ: 204.6, 164.8, 141.1, 140.6, 132.0, 131.5, 129.2, 129.0, 128.3, 127.9, 127.8, 127.7, 127.1, 126.6, 107.8, 100.6, 67.1, 67.0, 47.8, 42.6. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{26}H_{24}NO_2^+$ 382.1807; Found 382.1793.

*4-(4-Morpholino-4-oxo-3-phenylbuta-1,2-dien-1-yl)phenyl acetate (**3ga**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 6:1) as a yellow liquid (47.9 mg, 66% yield). ^1H NMR (500 MHz, $CDCl_3$) δ 7.43−7.33 (m, 6H), 7.28 (t, J = 7.1 Hz, 1H), 7.09−7.04 (m, 2H), 6.74 (s, 1H), 3.86−3.79 (m, 1H), 3.73 (d, J = 4.9 Hz, 3H), 3.63−3.50 (m, 4H), 2.30 (s, 3H). $^{13}C\{^1H\}$ NMR (126 MHz, $CDCl_3$) δ: 204.3, 169.5, 164.7, 150.5, 131.9, 130.3, 129.2, 128.5, 128.4, 126.6, 122.4, 107.9, 100.1, 67.04, 66.95, 47.8, 42.5, 21.3. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{22}H_{22}NO_4^+$ 364.1547; Found 364.1542.

*4-(4-Fluorophenyl)-1-morpholino-2-phenylbuta-2,3-dien-1-one (**3ha**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a yellow solid (56.2 mg, 87% yield, mp 100.7−101.8 °C). ^1H NMR (500 MHz, $CDCl_3$) δ 7.43−7.26 (m, 7H), 7.06−6.99 (m, 2H), 6.73 (s, 1H), 3.87−3.80 (m, 1H), 3.78−3.68 (m, 3H), 3.58 (dd, J = 9.0, 5.0 Hz, 3H), 3.52 (dt, J = 8.5, 5.5 Hz, 1H). $^{13}C\{^1H\}$ NMR (126 MHz, $CDCl_3$) δ 204.0 (d, J = 2.52 Hz), 164.7, 162.6 (d, J = 248.22 Hz), 131.9, 129.2, 129.1 (d, J = 7.56 Hz), 128.6 (d, J = 2.52 Hz), 128.4, 126.5, 116.1 (d, J = 22.68 Hz), 107.9, 99.9, 67.0, 66.9, 47.7, 42.5. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{20}H_{19}FNO_2^+$ 324.1397; Found 324.1386.

*4-(4-Chlorophenyl)-1-morpholino-2-phenylbuta-2,3-dien-1-one (**3ia**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a yellow solid (60.3 mg, 89% yield, mp 160.1−161.3 °C). ^1H NMR (500 MHz, $CDCl_3$) δ 7.40−7.34 (m, 4H), 7.31−7.26 (m, 5H), 6.72 (s, 1H), 3.87−3.80 (m, 1H), 3.77−3.69 (m, 3H), 3.59 (d, J = 3.9 Hz, 3H), 3.53 (dd, J = 11.7, 5.2 Hz, 1H). $^{13}C\{^1H\}$ NMR (126 MHz, $CDCl_3$) δ: 204.3, 164.5, 133.9, 131.7, 131.1, 129.3, 129.2, 128.7, 128.5, 126.5, 108.1, 100.0, 67.0, 66.9, 47.8, 42.5. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{20}H_{19}ClNO_2^+$ 340.1101; Found 340.1096.

*4-(4-Bromophenyl)-1-morpholino-2-phenylbuta-2,3-dien-1-one (**3ja**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a white solid (62.0 mg, 81% yield, mp 143.6−145.5 °C). ^1H NMR (500 MHz, $CDCl_3$) δ 7.45 (d, J = 8.4 Hz, 2H), 7.41−7.34 (m, 4H), 7.32−7.27 (m, 1H), 7.22 (d, J = 8.4 Hz, 2H), 6.70 (s, 1H), 3.83 (td, J = 10.0, 5.5 Hz, 1H), 3.78−3.70 (m, 3H), 3.59 (s, 3H), 3.53 (dd, J = 12.1, 5.6 Hz, 1H). $^{13}C\{^1H\}$ NMR (126 MHz, $CDCl_3$) δ: 204.4, 164.5, 132.2, 131.63, 131.60, 129.2, 129.0, 128.5, 126.5, 122.0, 108.2, 100.1, 67.0, 66.9, 47.8, 42.5. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{20}H_{19}BrNO_2^+$ 384.0596; Found 384.0587.

*1-Morpholino-2-phenyl-4-(pyren-1-yl)buta-2,3-dien-1-one (**3ka**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 6:1) as a yellow liquid (70.4 mg, 82% yield). ^1H NMR (500 MHz, $CDCl_3$) δ 8.42 (d, J = 9.3 Hz, 1H), 8.20 (dd, J = 7.6, 2.9 Hz, 2H), 8.14 (d, J = 9.3 Hz, 1H), 8.11 (s, 2H), 8.08 (d, J = 8.9 Hz, 1H), 8.04 (d, J = 2.9 Hz, 1H), 8.02 (t, J = 4.6 Hz, 1H), 7.79 (s, 1H), 7.57−7.52 (m, 2H), 7.42 (dd, J = 10.5, 4.9 Hz, 2H), 7.33 (t, J = 7.4 Hz, 1H), 3.92−3.85 (m, 1H), 3.79−3.70 (m, 3H), 3.66 (t, J = 4.5 Hz, 2H), 3.56−3.43 (m, 2H). $^{13}C\{^1H\}$ NMR (126 MHz, $CDCl_3$) δ 206.3, 164.9, 132.1, 131.5, 131.3, 130.9, 129.3, 128.34, 128.29, 127.8, 127.5, 126.6, 126.3, 126.1, 125.9, 125.8, 125.5, 125.4, 125.3, 124.9, 122.7, 107.2, 98.2, 67.0, 66.9, 47.8, 42.6. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{30}H_{24}NO_2^+$ 430.1808; Found 430.1792.

*4-(2,3-Dihydrobenzofuran-5-yl)-1-morpholino-2-phenylbuta-2,3-dien-1-one (**3la**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a white solid (61.1 mg, 88% yield, mp 123.5−125.3 °C). ^1H NMR (500 MHz, $CDCl_3$) δ 7.41 (d, J = 7.4 Hz, 2H), 7.35 (t, J = 7.7 Hz, 2H), 7.27 (t, J = 7.3 Hz, 1H), 7.23 (s, 1H), 7.08 (dd, J = 8.2, 1.4 Hz, 1H), 6.74 (d, J = 8.2 Hz, 1H), 6.71 (s, 1H), 4.57 (t, J = 8.7 Hz, 2H), 3.88−3.81 (m, 1H), 3.78−3.68 (m, 3H), 3.65−3.57 (m, 3H), 3.54 (dd, J = 12.8, 7.7 Hz, 1H), 3.17 (t, J = 8.7 Hz, 2H). $^{13}C\{^1H\}$ NMR (126 MHz, $CDCl_3$) δ: 203.7, 165.1, 160.4, 132.4, 129.1, 128.3, 128.1, 127.7, 126.5, 124.5, 124.0, 109.7, 107.5, 100.8, 71.7, 67.0, 66.9, 47.8, 42.5, 29.6. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{22}H_{22}NO_3^+$ 348.1598; Found 348.1587.

*4-(Furan-2-yl)-1-morpholino-2-phenylbuta-2,3-dien-1-one (**3ma**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a yellow liquid (49.0 mg,83% yield). ^1H NMR (500 MHz, $CDCl_3$) δ 7.41 (d, J = 7.7 Hz, 2H), 7.35 (t, J = 7.6 Hz, 3H), 7.29 (d, J = 7.4 Hz, 1H), 6.71 (s, 1H), 6.43 (d, J = 1.9 Hz, 1H), 6.37 (d, J = 3.3 Hz, 1H), 3.85 (ddd, J = 19.4, 9.0, 4.5 Hz, 2H), 3.75 (dd, J = 13.2, 8.5 Hz, 1H), 3.70 (t, J = 8.2 Hz, 2H), 3.66−3.60 (m, 1H), 3.58−3.51 (m, 2H). $^{13}C\{^1H\}$ NMR (126 MHz, $CDCl_3$) δ: 203.7, 164.7, 146.6, 143.1, 131.9, 129.1, 128.4, 126.8, 112.0, 109.8, 107.8, 91.0, 67.1, 66.9, 47.8, 42.6. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{18}H_{18}NO_3^+$ 296.1289; Found 296.1277.

*4-(Benzo[b]thiophen-3-yl)-1-morpholino-2-phenylbuta-2,3-dien-1-one (**3na**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 6:1 to 4:1) as a yellow solid (64.3 mg, 89% yield, mp 116.5−118.0 °C). ^1H NMR (500 MHz, $CDCl_3$) δ 8.09 (dd, J = 6.2, 2.7 Hz, 1H), 7.86 (dd, J = 6.1, 2.6 Hz, 1H), 7.49−7.44 (m, 3H), 7.41−7.35 (m, 4H), 7.30 (t, J = 7.3 Hz, 1H), 7.10 (s, 1H), 3.90−3.83 (m, 1H), 3.79−3.69 (m, 3H), 3.60 (d, J = 4.6 Hz, 2H), 3.58−3.47 (m, 2H). $^{13}C\{^1H\}$ NMR (126 MHz, $CDCl_3$) δ: 204.2, 164.8, 140.7, 137.0, 132.3, 129.2, 128.3, 127.4, 126.7, 125.7, 125.0, 123.0, 122.6, 106.7, 94.0, 67.0, 66.9, 47.6, 42.5. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{22}H_{20}NO_2S^+$ 362.1213; Found 362.1201.

*1-Morpholino-2,4-diphenylpenta-2,3-dien-1-one (**3oa**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a white solid (55.5 mg, 87% yield, mp 121.7−122.5 °C). ^1H NMR (500 MHz, $CDCl_3$) δ 7.45 (d, J = 7.9 Hz, 2H), 7.40 (d, J = 7.8 Hz, 2H), 7.35 (td, J = 7.6, 1.3 Hz, 4H), 7.27 (t, J = 7.2 Hz, 2H), 3.82 (td, J = 9.8, 5.8 Hz, 1H), 3.73 (dd, J = 11.2, 5.9 Hz, 3H), 3.64−3.43 (m, 4H), 2.28 (s, 3H). $^{13}C\{^1H\}$ NMR (126 MHz, $CDCl_3$) δ: 203.2, 165.4, 135.5, 132.8, 129.1, 128.8, 128.0, 126.5, 126.2, 107.6, 105.6, 67.1, 67.0, 47.8, 42.5, 16.6. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{21}H_{22}NO_2^+$ 320.1648; Found 320.1641.

*4-([1,1′-Biphenyl]-4-yl)-1-morpholino-2-phenylpenta-2,3-dien-1-one (**3pa**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a white solid (67.2 mg, 85% yield, mp 135.8−137.7 °C). ^1H NMR (500 MHz, $CDCl_3$) δ 7.61−7.57 (m, 4H), 7.55−7.51 (m, 2H), 7.47−7.40 (m, 4H), 7.39−7.33 (m, 3H), 7.30−7.26 (m, 1H), 3.84 (dt, J = 12.4, 6.6 Hz, 1H), 3.78−3.70 (m, 3H), 3.65−3.60 (m, 2H), 3.59−3.49 (m, 2H), 2.31 (s, 3H). $^{13}C\{^1H\}$ NMR (126 MHz, $CDCl_3$) δ: 203.4, 165.3, 140.8, 140.6, 134.4, 132.7, 129.1, 129.0, 128.0, 127.6, 127.5, 127.1, 126.6, 126.5, 107.4, 105.8, 67.1, 67.0, 47.8, 42.5, 16.6. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{27}H_{26}NO_2^+$ 396.1964; Found 396.1949.

*1-Morpholino-4-(naphthalen-2-yl)-2-phenylpenta-2,3-dien-1-one (**3qa**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a white solid (68.6 mg, 93% yield, mp 131.8−133.5 °C). ^1H NMR (500 MHz, $CDCl_3$) δ 7.86−7.84 (m, 2H), 7.82−7.79 (m, 1H), 7.77 (d, J = 8.6 Hz, 1H), 7.61 (dd, J = 8.6, 1.7 Hz, 1H), 7.52−7.46 (m, 2H), 7.46−7.43 (m, 2H), 7.38 (t, J = 7.7 Hz, 2H), 7.29 (t, J = 7.3 Hz, 1H), 3.91−3.48 (m, 8H), 2.40 (s, 3H). $^{13}C\{^1H\}$ NMR (126 MHz, $CDCl_3$) δ: 203.8, 165.3, 133.6, 133.0, 132.9, 132.8, 129.1, 128.3, 128.2, 128.0, 127.8, 126.5, 126.3, 125.0, 124.3, 107.8, 105.9, 67.1, 66.9, 47.8, 42.5, 16.6.

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HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{25}H_{24}NO_2^+$ 370.1806; Found 370.1797.

*4-(4-Methoxyphenyl)-1-morpholino-2,5-diphenylpenta-2,3-dien-1-one (**3ra**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a white solid (67.2 mg, 79% yield, mp 126.0−127.7 °C). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.43 (d, J = 8.8 Hz, 2H), 7.36−7.28 (m, 6H), 7.24 (t, J = 7.6 Hz, 3H), 7.17 (t, J = 7.2 Hz, 1H), 6.86 (d, J = 8.8 Hz, 2H), 4.03 (d, J = 15.9 Hz, 1H), 3.85 (d, J = 15.9 Hz, 1H), 3.78 (d, J = 8.9 Hz, 4H), 3.70−3.56 (m, 3H), 3.26 (d, J = 16.4 Hz, 2H), 3.03−2.91 (m, 2H). ^{13}C{^{1}H} NMR (126 MHz, $CDCl_3$) δ: 203.7, 165.4, 159.5, 138.7, 132.7, 129.2, 129.1, 128.6, 127.9, 127.8, 127.0, 126.7, 126.2, 114.3, 112.5, 107.7, 67.1, 66.9, 55.4, 47.0, 42.3, 37.3. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{28}H_{28}NO_3^+$ 426.2070; Found 426.2056.

*3-(3,4-Dihydronaphthalen-1(2H)-ylidene)-1-morpholino-2-phenylprop-2-en-1-one (**3sa**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 10:1 to 4:1) as a white solid (55.9 mg, 81% yield, mp 132.9−134.5 °C). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.45 (d, J = 7.6 Hz, 1H), 7.41−7.38 (m, 2H), 7.34 (t, J = 7.7 Hz, 2H), 7.29−7.24 (m, 1H), 7.20−7.14 (m, 2H), 7.13−7.09 (m, 1H), 3.82 (dd, J = 14.6, 7.4 Hz, 1H), 3.79−3.71 (m, 3H), 3.65 (d, J = 3.5 Hz, 2H), 3.56 (d, J = 4.2 Hz, 2H), 2.88 (t, J = 6.2 Hz, 2H), 2.80−2.64 (m, 2H), 2.04−1.94 (m, 2H). ^{13}C{^{1}H} NMR (126 MHz, $CDCl_3$) δ: 201.0, 165.5, 137.2, 133.1, 129.8, 129.7, 129.1, 127.92, 127.88, 127.5, 126.52, 126.49, 108.7, 106.6, 67.2, 67.0, 47.9, 42.5, 29.9, 28.4, 23.1. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{23}H_{24}NO_2^+$ 346.1805; Found 346.1793.

*4-(4-Methoxyphenyl)-1-morpholino-2-(p-tolyl)buta-2,3-dien-1-one (**3ta**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a white solid (62.9 mg, 90% yield, mp 114.0−115.3 °C). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.28 (t, J = 8.9 Hz, 4H), 7.16 (d, J = 8.0 Hz, 2H), 6.86 (d, J = 8.7 Hz, 2H), 6.71 (s, 1H), 3.87−3.78 (m, 4H), 3.76−3.68 (m, 3H), 3.63−3.48 (m, 4H), 2.34 (s, 3H). ^{13}C{^{1}H} NMR (126 MHz, $CDCl_3$) δ: 203.5, 165.2, 159.6, 138.1, 129.8, 129.3, 128.7, 126.4, 124.9, 114.5, 107.4, 100.3, 67.1, 66.9, 55.5, 47.8, 42.5, 21.3. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{22}H_{24}NO_3^+$ 350.1754; Found 350.1746.

*2,4-Bis(4-methoxyphenyl)-1-morpholinobuta-2,3-dien-1-one (**3ua**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a yellow solid (66.4 mg, 91% yield, mp 118.9−120.3 °C). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.33 (d, J = 8.8 Hz, 2H), 7.27 (d, J = 8.7 Hz, 2H), 6.87 (t, J = 8.4 Hz, 4H), 6.70 (s, 1H), 3.82 (d, J = 19.4 Hz, 7H), 3.77−3.69 (m, 3H), 3.63−3.49 (m, 4H). ^{13}C{^{1}H} NMR (126 MHz, $CDCl_3$) δ: 203.1, 165.2, 159.6, 128.6, 127.8, 125.0, 124.4, 114.6, 114.5, 107.1, 100.3, 67.1, 66.9, 55.5, 47.8, 42.5. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{22}H_{24}NO_4^+$ 366.1703; Found 366.1691.

*2-([1,1′-Biphenyl]-4-yl)-4-(4-methoxyphenyl)-1-morpholinobuta-2,3-dien-1-one (**3va**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a yellow solid (64.9 mg, 79% yield, mp 139.6−140.9 °C). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.59 (d, J = 8.3 Hz, 4H), 7.49 (d, J = 8.3 Hz, 2H), 7.44 (t, J = 7.6 Hz, 2H), 7.35 (t, J = 7.4 Hz, 1H), 7.30 (d, J = 8.7 Hz, 2H), 6.89 (d, J = 8.7 Hz, 2H), 6.77 (s, 1H), 3.90−3.84 (m, 1H), 3.82 (s, 3H), 3.79−3.71 (m, 3H), 3.67−3.52 (m, 4H). ^{13}C{^{1}H} NMR (126 MHz, $CDCl_3$) δ: 204.1, 165.0, 159.8, 141.0, 140.6, 131.3, 129.0, 128.8, 127.8, 127.6, 127.1, 127.0, 124.7, 114.6, 107.3, 100.5, 67.1, 67.0, 55.5, 47.8, 42.5. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{27}H_{26}NO_3^+$ 412.1913; Found 412.1898.

*4-(4-Methoxyphenyl)-1-morpholino-2-(naphthalen-2-yl)buta-2,3-dien-1-one (**3wa**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a yellow solid (64.0 mg, 83% yield, mp 137.2−138.7 °C). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.80 (dt, J = 9.6, 3.0 Hz, 4H), 7.55 (dd, J = 8.6, 1.7 Hz, 1H), 7.51−7.44 (m, 2H), 7.34−7.29 (m, 2H), 6.93−6.86 (m, 2H), 6.81 (s, 1H), 3.89 (dt, J = 10.6, 6.3 Hz, 1H), 3.84−3.74 (m, 6H), 3.69−3.49 (m, 4H). ^{13}C{^{1}H} NMR (126 MHz, $CDCl_3$) δ: 204.4, 165.1, 159.8, 133.7, 133.1, 129.6, 128.9, 128.8, 128.3, 127.8, 126.6, 126.5, 125.3, 124.7, 124.4, 114.6, 107.9, 100.7, 67.1, 67.0, 55.5, 47.9, 42.6. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{25}H_{24}NO_3^+$ 386.1755; Found 386.1740.

*4-(4-Methoxyphenyl)-1-morpholino-2-(thiophen-3-yl)buta-2,3-dien-1-one (**3xa**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 10:1 to 6:1) as a yellow liquid (55.3 mg, 81% yield). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.32 (dd, J = 1.7, 0.9 Hz, 1H), 7.29 (dd, J = 5.0, 2.9 Hz, 1H), 7.28−7.26 (m, 1H), 7.25 (d, J = 2.8 Hz, 1H), 7.12 (dd, J = 5.0, 1.3 Hz, 1H), 6.89−6.85 (m, 2H), 6.66 (s, 1H), 3.81 (s, 4H), 3.71 (t, J = 12.2 Hz, 3H), 3.64−3.49 (m, 4H). ^{13}C{^{1}H} NMR (126 MHz, $CDCl_3$) δ: 204.0, 165.0, 159.7, 133.3, 128.8, 126.5, 126.4, 124.7, 122.2, 114.6, 103.0, 99.6, 67.0, 67.0, 55.5, 47.9, 42.6. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{19}H_{20}NO_3S^+$ 342.1161; Found 342.1153.

*4-(6,6-Dimethylbicyclo[3.1.1]hept-2-en-2-yl)-1-morpholino-2-phenylbuta-2,3-dien-1-one (**3ya**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a yellow liquid (57.3 mg, 82% yield). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.36−7.21 (m, 5H), 6.46 (d, J = 2.4 Hz, 1H), 5.70 (s, 1H), 3.83−3.63 (m, 4H), 3.62−3.47 (m, 4H), 2.45−2.32 (m, 3H), 2.28 (t, J = 5.5 Hz, 1H), 2.13 (s, 1H), 1.25 (d, J = 6.3 Hz, 4H), 0.87 (d, J = 19.2 Hz, 3H). ^{13}C{^{1}H} NMR (126 MHz, $CDCl_3$) δ 203.5, 203.3, 165.2, 140.9, 140.8, 132.8, 129.0, 127.88, 127.85, 126.3, 126.2, 124.6, 124.5, 107.1, 106.9, 102.8, 67.13, 67.06, 67.0, 47.7, 43.74, 43.71, 42.42, 42.35, 40.7, 40.7, 38.18, 38.16, 32.3, 32.2, 31.6, 31.4, 26.3, 21.14, 21.06. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{23}H_{28}NO_2^+$ 350.2118; Found 350.2108.

*4-(3-((5-(4-Fluorophenyl)thiophen-2-yl)methyl)-4-methylphenyl)-1-morpholino-2-phenylbuta-2,3-dien-1-one (**3za**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a yellow liquid (79.4 mg, 78% yield). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.44 (dd, J = 8.7, 5.3 Hz, 2H), 7.40 (d, J = 7.5 Hz, 2H), 7.34 (t, J = 7.6 Hz, 2H), 7.28 (d, J = 7.3 Hz, 1H), 7.18 (d, J = 2.8 Hz, 2H), 7.15 (d, J = 8.3 Hz, 1H), 7.01 (dd, J = 10.2, 7.1 Hz, 3H), 6.74 (s, 1H), 6.67 (d, J = 3.5 Hz, 1H), 4.11 (s, 2H), 3.84−3.67 (m, 4H), 3.59−3.43 (m, 4H), 2.33 (s, 3H). ^{13}C{^{1}H} NMR (126 MHz, $CDCl_3$) δ 204.3, 164.9, 162.3 (d, J = 239.4 Hz), 142.9, 141.8, 139.0, 136.7, 132.2, 131.3, 130.9 (d, J = 12.6 Hz), 130.5, 129.1, 128.4, 128.2, 127.26 (d, J = 12.6 Hz), 126.6, 126.3, 126.1, 122.8, 115.86 (d, J = 12.6 Hz), 107.6, 100.7, 67.0, 66.9, 47.8, 42.5, 34.1, 19.5. ^{19}F NMR (471 MHz, $CDCl_3$) δ −115.03. HRMS (ESI)m/z: $[M + H]^+$ Calcd for $C_{32}H_{29}FNO_2S^+$ 510.1906; Found 510.1894.

*4-(4-Chloro-3-(4-((tetrahydrofuran-3-yl)oxy)benzyl)phenyl)-1-morpholino-2-phenylbuta-2,3-dien-1-one (**3Aa**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a yellow liquid (77.3 mg, 75% yield). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.35 (d, J = 4.2 Hz, 4H), 7.32 (d, J = 8.2 Hz, 1H), 7.29 (dd, J = 8.6, 4.3 Hz, 1H), 7.16 (dd, J = 8.2, 2.0 Hz, 1H), 7.09−7.06 (m, 3H), 6.76 (d, J = 8.6 Hz, 2H), 6.66 (s, 1H), 4.84 (t, J = 5.0 Hz, 1H), 4.01 (d, J = 4.5 Hz, 2H), 3.99−3.92 (m, 3H), 3.88 (dt, J = 8.3, 4.2 Hz, 1H), 3.82−3.64 (m, 5H), 3.47 (s, 4H), 2.27−2.10 (m, 2H). ^{13}C{^{1}H} NMR (126 MHz, $CDCl_3$) δ: 204.4, 164.5, 156.1, 139.8, 133.8, 131.8, 131.5, 131.4, 130.2, 130.1, 129.7, 129.2, 128.4, 126.5, 115.5, 108.0, 100.0, 73.2, 73.2, 67.3, 67.0, 66.9, 47.7, 42.5, 38.3, 33.1. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{31}H_{31}ClNO_4^+$ 516.1944; Found 516.1929.

*13-Methyl-3-(1-morpholino-1-oxo-4-phenylbuta-2,3-dien-2-yl)-6,7,8,9,11,12,13,14,15,16-decahydro-17H-cyclopenta[a]-phenanthren-17-one (**3Ba**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a yellow solid (80.8 mg, 84% yield, mp 128.8−130.1 °C). ^{1}H NMR

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(500 MHz, $CDCl_3$) δ 7.33 (q, J = 7.8 Hz, 4H), 7.28 (d, J = 8.3 Hz, 1H), 7.24 (d, J = 1.3 Hz, 1H), 7.19 (d, J = 8.2 Hz, 1H), 7.13 (s, 1H), 6.73 (d, J = 1.1 Hz, 1H), 3.82 (dd, J = 14.0, 7.6 Hz, 1H), 3.74−3.69 (m, 2H), 3.67 (dd, J = 10.0, 5.1 Hz, 2H), 3.61 (d, J = 4.9 Hz, 2H), 3.58−3.48 (m, 2H), 3.47−3.42 (m, 1H), 2.92−2.86 (m, 2H), 2.50 (dd, J = 19.0, 8.6 Hz, 1H), 2.41 (dd, J = 13.0, 4.4 Hz, 1H), 2.29 (dd, J = 13.8, 7.1 Hz, 1H), 2.16 (t, J = 9.0 Hz, 1H), 2.09 (s, 1H), 1.98−1.92 (m, 1H), 1.65−1.55 (m, 2H), 1.54−1.47 (m, 3H), 0.90 (s, 3H). $^{13}C\{^1H\}$ NMR (126 MHz, $CDCl_3$) δ 204.2, 169.3, 164.9, 140.1, 137.4, 132.7, 129.3, 129.0, 128.1, 127.5, 126.9, 126.2, 124.0, 107.4, 100.7, 67.0, 67.0, 66.7, 50.6, 48.1, 46.8, 44.5, 42.5, 38.1, 35.9, 31.6, 29.5, 26.5, 21.7, 13.9. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{32}H_{36}NO_3^+$ 482.2698; Found 482.2682.

*4-(4-Methoxyphenyl)-2-phenyl-1-thiomorpholinobuta-2,3-dien-1-one (**3ab**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a yellow solid (63.1 mg, 90% yield, mp 121.3−122.7 °C). ^1H NMR (500 MHz, $CDCl_3$) δ 7.39 (d, J = 7.4 Hz, 2H), 7.34 (t, J = 7.6 Hz, 2H), 7.31−7.26 (m, 3H), 6.87 (d, J = 8.7 Hz, 2H), 6.74 (s, 1H), 4.18−4.11 (m, 1H), 3.99−3.88 (m, 2H), 3.86−3.78 (m, 4H), 2.75−2.63 (m, 2H), 2.51−2.40 (m, 2H). $^{13}C\{^1H\}$ NMR (126 MHz, $CDCl_3$) δ: 203.5, 165.2, 159.7, 132.3, 129.1, 128.7, 128.2, 126.5, 124.8, 114.6, 107.8, 100.5, 55.5, 50.0, 44.5, 28.1, 27.6. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{21}H_{22}NO_2S^+$ 352.1369; Found 352.1359.

*1-(4-Butylpiperazin-1-yl)-4-(4-methoxyphenyl)-2-phenylbuta-2,3-dien-1-one (**3ac**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a yellow liquid (68.6 mg, 88% yield). ^1H NMR (500 MHz, $CDCl_3$) δ 7.42−7.38 (m, 2H), 7.33 (dd, J = 10.4, 4.9 Hz, 2H), 7.29 (t, J = 2.3 Hz, 2H), 7.27−7.24 (m, 1H), 6.88−6.84 (m, 2H), 6.71 (s, 1H), 3.80 (s, 3H), 3.64−3.60 (m, 2H), 2.54−2.42 (m, 2H), 2.38−2.27 (m, 4H), 1.44 (dt, J = 9.5, 4.5 Hz, 2H), 1.35−1.21 (m, 4H), 0.90 (t, J = 7.3 Hz, 3H). $^{13}C\{^1H\}$ NMR (126 MHz, $CDCl_3$) δ: 203.7, 164.8, 159.6, 132.5, 129.0, 128.7, 128.0, 126.6, 125.0, 114.5, 107.9, 100.2, 58.3, 55.5, 53.7, 53.0, 47.4, 42.0, 29.0, 20.7, 14.1. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{25}H_{31}N_2O_2^+$ 391.2385; Found 391.2375.

*4-(4-Methoxyphenyl)-2-phenyl-1-(4-(pyrrolidin-1-yl)piperidin-1-yl)buta-2,3-dien-1-one (**3ad**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a yellow solid (69.2 mg, 86% yield, mp 130.1−131.6 °C). ^1H NMR (500 MHz, $CDCl_3$) δ 7.38 (d, J = 7.6 Hz, 2H), 7.32 (t, J = 7.6 Hz, 2H), 7.27 (d, J = 8.4 Hz, 2H), 7.23 (d, J = 7.2 Hz, 1H), 6.85 (dd, J = 8.5, 4.7 Hz, 2H), 6.69 (d, J = 5.6 Hz, 1H), 4.63 (dd, J = 18.4, 14.5 Hz, 1H), 4.16−4.05 (m, 1H), 3.79 (d, J = 3.2 Hz, 3H), 3.17−3.07 (m, 1H), 2.97−2.79 (m, 1H), 2.54 (d, J = 20.3 Hz, 4H), 2.21 (t, J = 10.2 Hz, 1H), 2.03−1.94 (m, 1H), 1.77 (dd, J = 6.4, 3.1 Hz, 4H), 1.59−1.39 (m, 2H), 1.37−1.25 (m, 1H). $^{13}C\{^1H\}$ NMR (126 MHz, $CDCl_3$) δ: 203.4, 164.6, 159.6, 132.6, 129.0, 128.7, 127.9, 126.5, 125.1, 114.5, 108.1, 100.2, 61.9, 55.5, 51.7, 51.6, 46.1, 46.1, 40.7, 32.2, 31.3, 23.3. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{26}H_{31}N_2O_2^+$ 403.2385; Found 403.2375.

*4-(4-Methoxyphenyl)-2-phenyl-1-(1,4-dioxa-8-azaspiro[4.5]-decan-8-yl)buta-2,3-dien-1-one (**3ae**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a yellow solid (68.2 mg, 87% yield, mp 129.5−131.1 °C). ^1H NMR (500 MHz, $CDCl_3$) δ 7.39 (d, J = 7.5 Hz, 2H), 7.33 (t, J = 7.6 Hz, 2H), 7.27 (t, J = 7.7 Hz, 3H), 6.86 (d, J = 8.7 Hz, 2H), 6.71 (s, 1H), 4.00−3.86 (m, 5H), 3.80 (m, 4H), 3.76−3.64 (m, 2H), 1.76 (t, J = 5.8 Hz, 2H), 1.57 (d, J = 7.6 Hz, 2H). $^{13}C\{^1H\}$ NMR (126 MHz, $CDCl_3$) δ: 203.4, 164.8, 159.6, 132.5, 129.1, 128.7, 128.0, 126.5, 124.9, 114.6, 107.9, 107.1, 100.3, 64.6, 55.5, 45.5, 40.0, 35.7, 34.9. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{24}H_{26}NO_4^+$ 392.1861; Found 392.1845.

*tert-Butyl-4-((4-(4-methoxyphenyl)-N-methyl-2-phenylbuta-2,3-dienamido)methyl)piperidine-1-carboxylate (**3af**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 6:1 to 4:1) as a white solid (77.3 mg, 81% yield, mp 141.2−142.9 °C). ^1H NMR (500 MHz, $CDCl_3$) δ 7.40 (d, J = 7.5 Hz, 2H), 7.33 (t, J = 7.6 Hz, 2H), 7.31−7.28 (m, 2H), 7.27 (d, J = 9.2 Hz, 1H), 6.86 (d, J = 8.7 Hz, 2H), 6.71 (s, 1H), 4.07 (s, 2H), 3.80 (d, J = 2.7 Hz, 3H), 3.30 (ddd, J = 21.0, 14.1, 7.5 Hz, 1H), 3.07 (d, J = 4.0 Hz, 3H), 2.58 (dd, J = 49.5, 31.2 Hz, 2H), 1.73 (ddd, J = 11.3, 10.7, 6.8 Hz, 1H), 1.63 (d, J = 12.7 Hz, 1H), 1.44 (d, J = 8.9 Hz, 11H), 1.29−1.16 (m, 1H), 1.03−0.84 (m, 1H). $^{13}C\{^1H\}$ NMR (126 MHz, $CDCl_3$) δ: 203.8, 166.5, 159.6, 154.9, 132.5, 129.0, 128.7, 128.0, 126.6, 124.9, 114.5, 108.4, 100.1, 79.5, 56.7, 55.5, 53.1, 38.1, 35.5, 34.7, 29.9, 28.6, 28.5. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{29}H_{37}N_2O_4^+$ 477.2755; Found 477.2741.

*4-(4-Methoxyphenyl)-N-methyl-N,2-diphenylbuta-2,3-dienamide (**3ag**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a yellow liquid (39.7 mg, 56% yield). ^1H NMR (500 MHz, $CDCl_3$) δ 7.41 (d, J = 7.6 Hz, 2H), 7.32 (t, J = 7.6 Hz, 2H), 7.27−7.22 (m, 4H), 7.10 (s, 2H), 6.79 (s, 2H), 6.72 (d, J = 8.1 Hz, 2H), 6.24 (s, 1H), 3.78 (s, 3H), 3.44 (s, 3H). $^{13}C\{^1H\}$ NMR (126 MHz, $CDCl_3$) δ: 206.0, 166.3, 159.3, 144.5, 133.5, 129.4, 128.8, 128.7, 127.7, 127.0, 126.8, 126.5, 124.6, 114.2, 108.7, 99.7, 55.4, 37.7. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{24}H_{22}NO_2^+$ 356.1649; Found 356.1644.

*5,6-Dimethoxy-2-((1-(4-(4-methoxyphenyl)-2-phenylbuta-2,3-dienoyl)piperidin-4-yl)methyl)-2,3-dihydro-1H-inden-1-one (**3ah**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 6:1 to 4:1) as a white solid (89.3 mg, 83% yield, mp 106.4−108.3 °C). ^1H NMR (500 MHz, $CDCl_3$) δ 7.39 (dd, J = 6.8, 4.9 Hz, 2H), 7.34 (t, J = 7.7 Hz, 2H), 7.31−7.24 (m, 3H), 7.15 (d, J = 4.6 Hz, 1H), 6.86 (dd, J = 8.4, 3.5 Hz, 3H), 6.71 (d, J = 2.8 Hz, 1H), 4.81−4.67 (m, 1H), 4.14 (d, J = 9.1 Hz, 1H), 3.96 (s, 3H), 3.90 (s, 3H), 3.80 (d, J = 2.2 Hz, 3H), 3.29−3.15 (m, 1H), 3.12−3.00 (m, 1H), 2.77 (dd, J = 13.8, 7.9 Hz, 1H), 2.71−2.56 (m, 2H), 1.92−1.65 (m, 5H), 1.29 (dd, J = 15.3, 5.5 Hz, 1H), 1.21−1.09 (m, 1H). $^{13}C\{^1H\}$ NMR (126 MHz, $CDCl_3$) δ: 207.5, 203.4, 164.7, 159.6, 155.7, 149.6, 148.7, 132.6, 129.0, 128.8, 128.7, 128.0, 126.5, 125.1, 114.5, 108.2, 107.5, 104.5, 100.2, 56.4, 56.2, 55.5, 47.7, 45.3, 42.3, 38.6, 35.0, 34.8, 33.6, 32.4. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{34}H_{36}NO_5^+$ 538.2597; Found 538.2578.

*1-(4-(2-Chlorodibenzo[b,f][1,4]oxazepin-11-yl)piperazin-1-yl)-4-(4-methoxyphenyl)-2-phenylbuta-2,3-dien-1-one (**3ai**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a yellow solid (86.4 mg, 77% yield, mp 141.1−142.5 °C). ^1H NMR (500 MHz, $CDCl_3$) δ 7.41 (dd, J = 11.7, 5.3 Hz, 3H), 7.35 (dd, J = 10.5, 4.9 Hz, 2H), 7.32−7.26 (m, 4H), 7.19 (d, J = 8.6 Hz, 1H), 7.15 (d, J = 7.5 Hz, 1H), 7.09 (dd, J = 11.9, 4.5 Hz, 2H), 7.04−7.00 (m, 1H), 6.90−6.86 (m, 2H), 6.75 (s, 1H), 3.90 (d, J = 4.6 Hz, 2H), 3.83−3.71 (m, 5H), 3.53 (d, J = 47.0 Hz, 4H). $^{13}C\{^1H\}$ NMR (126 MHz, $CDCl_3$) δ: 203.8, 165.2, 159.7, 159.5, 158.8, 151.9, 133.0, 132.4, 130.6, 129.1, 129.0, 128.8, 128.2, 127.2, 126.5, 126.0, 125.20, 125.19, 124.8, 124.7, 123.0, 120.3, 114.6, 107.8, 100.5, 55.5, 47.0, 41.8. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{34}H_{29}ClN_3O_3^+$ 562.1901; Found 562.1885.

*1-(4-(8-Chloro-5,6-dihydro-11H-benzo[5,6]cyclohepta[1,2-b]-pyridin-11-ylidene)piperidin-1-yl)-4-(4-methoxyphenyl)-2-phenylbuta-2,3-dien-1-one (**3aj**).* Following the general procedure, the title compound was purified by flash chromatography on neutral alumina eluting with petroleum ether/EtOAc (v/v = 8:1 to 4:1) as a yellow solid (91.5 mg, 82% yield, mp 116.9−118.5 °C). ^1H NMR (500 MHz, $CDCl_3$) δ 8.43−8.30 (m, 1H), 7.41 (dd, J = 18.5, 10.1 Hz, 3H), 7.32 (td, J = 7.7, 2.4 Hz, 2H), 7.28−7.21 (m, 3H), 7.18−7.04 (m, 4H), 6.89−6.81 (m, 2H), 6.75−6.61 (m, 1H), 4.30−4.14 (m, 1H), 3.90 (ddd, J = 22.5, 17.3, 8.8 Hz, 1H), 3.79 (dd, J = 6.1, 3.9 Hz, 3H), 3.39−3.24 (m, 4H), 2.87−2.73 (m, 2H), 2.62−2.16 (m, 4H). $^{13}C\{^1H\}$ NMR (126 MHz, $CDCl_3$) δ 203.4, 164.9, 159.6, 156.8, 146.8, 139.6, 137.8, 137.6, 136.9, 134.8, 133.5, 133.2, 132.5, 130.5, 129.0, 128.7, 128.0, 126.5, 126.4, 126.2, 125.0, 122.4, 114.5, 108.1,

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100.3, 55.4, 48.2, 48.1, 43.0, 31.7, 31.6, 30.7. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{36}H_{32}ClN_2O_2^+$ 559.2147; Found 559.2141.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.4c00977.

Experimental procedures and compound characterization (PDF)

FAIR data, including the primary NMR FID files, for compounds **3aa**−**3BA** and **3ab**−**3aj** (ZIP)

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Notes

The authors declare no competing financial interest.

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